

VASTRANGAM

MEDHAVA — PLAN OF ACTION

One business operating system. Any industry. One shared data core.

22 modules · 113 apps · 285 rules (86 enforced) · 113 tables · 46 product screens across 12 sectors · 19 tool capabilities · 6 industry packs. Every count in this line is read from `brand/site/modules.js`, `brand/site/rules.js`, `brand/site/shots.js`, `brand/site/tools.js`, `core/schema.postgres.sql` and `core/packs/` by `brand/site/mkcounts.js` — no figure here was typed from memory.

What this document is. The plan for building Medhava as a product that any business can adopt, from planning through execution. It is not the Vastrangam plan with the names changed. `PLAN_OF_ACTION.md` is that: one ethnic-wear group in Surat adopting this engine, with its own karigar payroll, its own three companies, its own migration off an accounting package. That document stays as it is, and it is the most useful thing in this repository — it is the worked example of the whole exercise, an entire industry implementation carried to the level of detail that proves the engine bends.

This document answers the different question: **how does the same engine serve a dental clinic and a freight forwarder and a dairy co-operative, without becoming a different program each time?**

PART I — WHAT MEDHAVA IS

M1 · ONE ENGINE, MANY TRADES

Medhava is a single application over one shared data core. A business event is entered once and flows everywhere it belongs — that law, and the cascades that follow from it, are set out in full in `PLAN_OF_ACTION.md` §A0 and are identical here, because it is the same engine. This document does not restate them.

What is specific to Medhava is the claim that the engine does not know which trade it is in. That claim is either true in the code or it is marketing, so here is where it is enforced:

The claim	What makes it true
No module assumes an industry	<code>brand/site/modules.js</code> is audited at build time; trade vocabulary in it fails the build
Trade words live in one place	The edition overlay may change wording only — <code>build.js</code> compares the structural shape before and after applying it and exits if a module number, an app name or an app count moved
The same structure ships twice already	Two editions build from one module list: MEDHAVA neutral, VASTRANGAM in one trade's own words
The number of companies is data	<code>core/tests/core.test.js</code> posts across a 10 × 10 company × channel grid, then runs 11 × 11 with no code changed

The sentence the whole product rests on. A clinic's appointment, a factory's work order, a law firm's matter, a contractor's site and a service firm's job are the same record with different words on it. Everything below is the work of making that true rather than clever.

M2 · THE INDUSTRY PACK ENGINE

An earlier version of this document said, in this place, that the industry pack was **specified, not built** — that a third trade meant someone writing a third overlay by hand, and that this did not scale to a product. That was the honest state of it then. It is built now, and the paragraphs below say what was built, what it refuses, and which trades it was pointed at first and why.

M2.1 · Which trades actually run this software

The pack order was not chosen by taste. It was chosen by looking up who actually buys ERP, order management and warehouse management, and building for the largest first. Published estimates differ between research houses — sometimes considerably, because "share of users" and "share of revenue" are different questions — so the figures below are attributed, and the ranking rather than any single number is what the build order rests on.

ERP — who the users are

Industry	Share of ERP users	Note
Manufacturing	~21% of users; ~32% of market revenue	The largest block on every measure. Cited elsewhere as 19.7% of revenue — the spread is why the ranking, not the decimal, is what is used
Banking, financial services, insurance	~16%	Heavily served by specialist systems rather than general ERP
Professional and financial services	13.86%	The second-largest block, and the one with no stock at all
Distribution and wholesale	9.90%	Credit and movement, not production
Healthcare	4.95%	Small today — and the fastest-growing, at 22.37% CAGR to 2030
Construction	1.98%	
Education	0.99%	

Finance and insurance, manufacturing and public administration together account for roughly 60% of total ERP spend.

WMS — who the users are

End user	Share	Note
Retail and e-commerce	~28%, about \$1.37bn in 2025, 10.1% CAGR	The largest WMS segment
Manufacturing	23.6–30.22% depending on the source	
Healthcare and pharmaceuticals	~10%	The fastest-growing, at 12.2% CAGR

The whole WMS market is put at \$3.4bn–\$5.92bn in 2025, again depending on who is counting and what they count as WMS.

OMS and third-party logistics — which markets are served

Market served by 3PLs	Share
Transportation	90%
Manufacturing	83%
Retail	83%
E-commerce	68%
Wholesale	67% (down from 83% the previous year)

And among the technology services those providers offer, **order management ranks first** — ahead of visibility (86%), transport management (84%), optimisation (77%) and ERP integration (75%). That is the finding that

matters most for Medhava's shape: for a large part of this market the order, not the ledger, is the system of record.

Sources, read 2026-08-23: Grand View Research, Mordor Intelligence, HG Insights, MarketsandMarkets, Manufacturing Lead Generation, Inbound Logistics (3PL Perspectives), Electro IQ, openpr, NetSuite. These are market-research estimates, not measurements; they are used here to order a build queue, which is what they are good for, and not quoted anywhere in the product as fact.

M2.2 · What a pack is

An industry pack is a row of configuration, loaded as data, that carries:

- **vocabulary** — what this trade calls an order, a customer, a unit of work, a stage
- **stages** — the pipeline a job moves through, named and ordered by the trade
- **fields** — the extra attributes this trade needs on a record, and their types
- **documents** — the papers it issues, and what has to be on them
- **rule switches** — which discretionary rules apply, and at what thresholds
- **starting data** — a chart of accounts, roles, units of measure

The engine reads a pack the way it already reads companies and channels: as rows. Nothing in `modules.js`, `core/` or the schema changes when a fourteenth trade is added — which is exactly the property the 10×10 test already proves for companies, applied to trades.

M2.3 · What a pack may never do

A configuration file that can do anything is not configuration, it is a hole. Six refusals are enforced in `core/packs.js` and each one is a named test in `core/tests/packs.test.js`:

A pack may not	Because
contain executable code, at any depth	the moment a pack can run code, adding a trade is a code change again and the whole guarantee is worthless
invent a concept the engine does not have	"vocabulary" would quietly become a place to put anything, and the screens would carry a name for something that does not exist
add a field to a table that does not exist	the shape of the database would move outside the reach of the schema test that guards it
declare money as a plain number	the entire schema was built to keep floating-point rupees out; a pack is not a side door
switch off an immutable rule	a trade may call an invoice a fee note; it may not decide its audit trail is optional
be applied in part	a half-loaded trade is a system whose vocabulary and rules disagree, with no way to tell which half is live

The rule ids marked immutable in `core/packs.js` cover company scoping, the audit trail, the posting rules, group elimination and roster privacy — and a test asserts that every one of them really exists in the rulebook, so the protection cannot silently guard nothing.

One default is worth stating on its own, because getting it backwards would be invisible: **a rule a pack never mentions is ON**. The rulebook is the default and a pack is an exception list, never a permission list. The other way round, every rule added after a pack was written would silently apply to nobody using it.

M2.4 · Which packs ship

Built in the order the research ranks them:

Rank	Pack	Sector	What it is really for
1	manufacturing	Manufacturing	The largest ERP user base, and the vocabulary furthest from a service firm's
2	wholesale-distribution	Distribution	Credit, not production, is the thing the software has to hold
3	retail-ecommerce	Retail	The largest WMS segment; returns and settlement are the hard part
4	professional-services	Services	No stock at all — the hardest case for the neutrality claim
5	healthcare-clinic	Healthcare	Small now, fastest-growing on every measure
6	logistics-3pl	Logistics	The most-served 3PL market, where the order <i>is</i> the product

M2.5 · The gate, and what it proves

Phase 2's gate was written before the engine was: *a third trade is added without writing code*.

`core/tests/packs.test.js` §4 runs it, and runs it harder than the phase asked. During the test run it invents a **commercial laundry** — a trade that appears nowhere in this repository, in no pack, in no module, in no rule — hands the engine a JSON string, and then requires the whole system to answer in that trade's words: an order reads as a docket, a work order as a wash load, a customer as an account. Its pipeline resolves ordered and terminating, its extra fields land on real tables, its rule switches resolve against the real rulebook, and it is refused the audit trail exactly as the six shipped packs are.

A last assertion closes the loop: **** `core/packs.js` may not contain a single trade word.** Not laundry, not clinic, not freight, not dealer, not matter, not patient. If the engine ever learns a trade, the test fails — because an engine that knows one trade's words has an opinion about which trades are normal, and that is the failure this whole section exists to prevent.

`node core/tests/packs.test.js` → **every check passes, 0 failures.**

M3 · TENANCY — WHAT IS A ROW AND WHAT IS CODE

Concept	Row or code	Consequence
Tenant (a customer of Medhava)	row	Onboarding a business is data entry, not a deployment
Company inside a tenant	row	A group with four companies is four rows, consolidated with inter-company trade eliminated
Channel	row	A new marketplace is a row; rule R15.1 makes discovery automatic
Location, stage, role	row	A warehouse, a production stage and a job title are all configuration
Industry pack	row	A trade is configuration, not a fork — <code>core/packs.js</code> , six packs shipped, a seventh added during the test run
The 22 modules	code	The structure is the product; it is the same for everyone
The rulebook	code	Which apply is configurable; what they refuse is not — and 22 of them cannot be switched off by any pack

Isolation. Every business table carries `company_id` and a row-level security policy carrying both `USING` and `WITH CHECK`, so a read and a write are separately prevented from crossing a boundary. `core/tests/schema.test.js` fails the build if any company-scoped table lacks one. Cross-tenant isolation is the same mechanism one level up and is the single highest-risk item in this plan — a bug there is not a defect, it is an incident.

M4 · THE 22 MODULES, READ FROM TWELVE TRADES

The module list is in `PLAN_OF_ACTION.md` Part II in full, with every app and every rule. It is not repeated here. What is here is the reading that makes it industry-neutral — the same module, seen from a different trade:

Module	Manufacturing	Professional services	Healthcare	Hospitality
02 Design & Sampling	part revision	matter scoping	protocol	recipe development
05 Sales	order book	engagement letter	appointment book	cover forecast
06 Planning	material requirement	resourcing	rota and stock	purchase plan
08 Manufacturing	work order, stages	—	—	batch and prep
09 Quality	inspection	file review	licence and calibration	food safety log
10 Warehouse	pick and pack	—	consumables	store issue
16 HR	shift and piece-rate	timesheets, chargeable	sessions and locums	rota and casuals
20 Projects	improvement work	matters	care pathways	openings

The landing page carries every one of the product screens counted at the head of this document, across all twelve sectors, doing exactly this — the same module rendered with a machine shop's figures and a clinic's figures, one under the other. That is the argument made where it can be checked rather than believed.

PART II — EXECUTION

M5 · THE STACK, AND WHAT IT COSTS

The full tools register is `brand/site/tools.js`, gated by `brand/site/checktools.js`, which fails the build if any paid tool does not name **both** the free option it replaces and the concrete condition that forces the upgrade. "When we grow" is rejected by the checker; a number or a named event passes.

19 capabilities. Only three have no free path in existence:

Capability	Why there is no free option	What it costs
WhatsApp Business messaging	Meta requires an approved provider; there is no free tier to outgrow	~₹1,500–3,000/month plus per-conversation charges
SMS	Carrier access is the cost	~₹0.15–0.25 per message
Generated imagery	Free software, but it needs a graphics card	A GPU, or a hosted API per image

Everything else runs on free tiers or free software until a stated trigger fires: Postgres, self-hosted automation, a free hosting tier, local AI, the browser's own speech synthesis, ffmpeg and Chromium for media, UPI for payments, CSV for every integration, and a progressive web app instead of a store listing. **A business can run the whole system on zero rupees of software licence** and pay only for the three lines above, and only when it reaches them.

The Provider Router in Module 01 puts a spend ceiling in front of every paid call and refuses past it rather than warning — so the AI line in particular cannot quietly become the largest one.

M6 · THE EIGHT PHASES

The gate is absolute: **Phase N+1 does not start until Phase N's tests pass.** A phase is done when its stated result is reproduced, not when its code is written.

Phase	What gets built	Done when
0 · Setup	Environments, CI, monitoring, the schema loaded	A commit deploys and a user logs in
1 · Foundation & tenancy	Tenants, companies, roles, RLS, masters, the SKU/item model	Two tenants exist and neither can read a single row of the other, proved by a test that tries
2 · The industry pack engine 	Vocabulary, stages, fields, documents and starting data as rows	Done. A seventh trade is added without writing code , during the test run, from a JSON string — <code>core/tests/packs.test.js</code> §4
3 · Core operations	Inventory, procurement, production, quality, warehouse	Three trades run the same operations screens on their own vocabulary
4 · Commerce & channels	Sales all channels, OMS, returns, logistics, settlement	A week of orders across channels, settled and reconciled
5 · Finance	Double-entry, tax, banking, period locks	A month closes: trial balance ties and returns generate from vouchers
6 · The AI layer	Assistant, chatbot, agents, guardrails, content	An assistant answer matches the books; an agent asked to move money stops
7 · Onboarding & self-serve	Sign-up, pack selection, import, go-live	A business onboards itself in a day without a call

Phase 2 is the one that matters, and it is the one that is finished. The gate was written before the engine was — *a new trade must be addable without a developer* — and it is now run on every test pass against a trade nobody designed for. Phase 3 onwards is built on top of a claim that has been checked rather than one that was argued past.

M7 · ONBOARDING A BUSINESS IN A DAY

For a business with no system to migrate from — which is most of the target market — the sixty-day parallel run in the Vastrangam plan is the wrong shape entirely. The sequence is:

1. **Sign up, pick a trade.** The industry pack loads vocabulary, stages, documents and a chart of accounts. Nothing is blank.

2. **Name the company.** One row. A group adds more rows now or later.
3. **Import what exists** — customers, suppliers, items, opening stock — from a spreadsheet, with a validation report **before** anything commits. Errors are shown as rows to fix, never silently skipped.
4. **Opening balances**, or none at all if the business is starting fresh.
5. **Invite people, set roles.** Permissions are per company per role from the first minute.
6. **Do one real transaction end to end** — one sale, invoiced, stock moved, posted — and click the dashboard figure down to the voucher. That is the go-live test, and it takes a minute.

For a business migrating from an existing system, the discipline in [PLAN_OF_ACTION.md](#) Part IV E5 applies unchanged: export, clean, load, opening balances, parallel run, and a decommission criterion agreed in advance rather than argued at the end.

M8 · SECURITY AND COMPLIANCE

Row-level security in the database and permission checks in API middleware — one layer is one mistake away from a cross-tenant read. Personal data is exportable and erasable on request, with consent and retention tracked as the two separate clocks they are. Card data never reaches this system; the gateway's own secured field takes it, so there is no scope to protect. The audit trail has no off switch, and R16.22 keeps individual pay and personal attributes out of any document that leaves the building.

M9 · PERFORMANCE

Operation	p95
Page load, cached / uncached	< 1 s / < 3 s
Live stock or availability query	< 500 ms
Document PDF	< 2 s
Channel order pull, per channel	< 60 s
Settlement import, 1,000 lines	< 30 s
Daily profit-and-loss refresh	< 10 s

The availability query is the one that decides whether people trust the system: it runs on every screen showing a quantity or a free slot, and one that takes two seconds is one people stop asking.

M10 · RISK REGISTER

Risk	Likelihood	Impact	Mitigation
Cross-tenant data leak	Low	Critical	RLS with USING and WITH CHECK, middleware checks, a test that actively attempts a cross-tenant read
The industry pack engine slips, and trades get hard-coded instead	High	Critical	Phase 2 gate: a third trade added with no code. Hold the gate
A vertical demands a genuine structural exception	Medium	High	Add it as a module or an app for everyone, never as a branch for one customer
Free tier terms change or a tier disappears	Medium	Medium	Every capability has a self-host route recorded; the register is dated
WhatsApp provider outage	Low	High	Adapter swap; SMS and in-app as fallback
Onboarding needs hand-holding, so it does not scale	High	Medium	The day-one sequence above is a tested path, not a document
AI spend runs away	Medium	Medium	The spend ceiling refuses rather than warns

M11 · SUCCESS METRICS

Measure	Target
A new trade supported without writing code	Yes, from Phase 2 onward — binary, not a percentage
Time for a business to onboard itself	Under a day, unattended
Cost to run for a business at pilot scale	₹0 in software licence
Rules enforced by a test	the enforced-of-total figure at the head of this document; up every build
Cross-tenant incidents	Zero, and this is the only metric with no acceptable non-zero value

M12 · RUNBOOKS

Daily — error and uptime check, failed integrations queue, onboarding funnel. **Weekly** — usage per tenant, slow queries, support themes that suggest a missing rule. **Monthly** — free-tier headroom against triggers, spend against ceilings, backup restore test. **Quarterly** — re-read the tools register against current published tiers, review which SPECIFIED rules became enforceable, and check whether any vertical has quietly grown an exception.

M13 · WHAT A CUSTOMER PAYS FOR

The free-first register is not only how Medhava is built — it is the shape of what it can offer. Because the system runs on free tiers and free software at pilot scale, a business can be given a genuinely working system before it pays anything, and the paid lines are the ones it would have had to pay anyway: messaging, marketplace access, payment processing, courier pickup.

The commercial decision — subscription, per-company, per-user, or usage — is not made in this document, because it is a business decision rather than an engineering one. What this document fixes is the constraint it must respect: **nothing in the architecture may make a plan cap technically necessary**. Companies, channels, users and trades are rows. If a plan limits them, that is a commercial choice stated in the pricing, and the software says so rather than pretending to a technical limit it does not have.

PART III — THE PROOF

The test of whether this is one system or twenty-two programs sharing a login is a single transaction followed end to end, and it is re-run at every module boundary. That walkthrough is in `PLAN_OF_ACTION.md` Part V and is identical here.

The test of whether it is *industry-neutral* is different, and it is this: **take the same transaction and run it in a trade nobody had in mind when the code was written**. That test used to be run by hand, by writing an overlay. It is now run by the machine: `core/tests/packs.test.js` invents a commercial laundry every time it executes, loads it from a JSON string, and requires the system to answer in that trade's words — while still refusing it the audit trail. The last assertion in that file is that `core/packs.js` contains no trade word at all, so the engine cannot quietly learn one trade and call it neutrality.

Every check passes, with the shipped packs and a seventh added at run time. That is the whole industry-neutrality claim, reduced to something that either passes or does not.

Counts in this document are read from `brand/site/modules.js`, `brand/site/rules.js`, `brand/site/shots.js`, `brand/site/tools.js` and `core/schema.postgres.sql` when it is generated. Product screens carry illustrative figures and are labelled with the trade they are drawn from; they are not case studies, and no business is named as a customer that is not one.
